CG	CLASS INDEX NO.
07S ___	

CANDIDATE NAME: ______________________________

TAMPINES JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008

BIOLOGY 9747/01
Paper 1 Multiple Choice

TUESDAY 26 AUGUST 2008 **12.30 – 1.45 PM (1 hour 15 minutes)**

Additional materials: Multiple Choice Answer Sheet

INSTRUCTIONS TO CANDIDATES

1. For **'Index Number'**, shade the **last 5 numbers** of your **NRIC** and shade letter **'A'** for local students and letter **'F'** for foreign students on the **Optical Answer Sheet** in the spaces designated.
2. Write in soft pencil.
3. Do not use staples, paper clips, highlighters, glue or correction fluid.

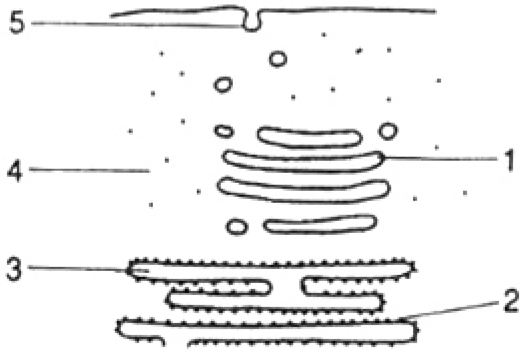
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculations may be used.

This question paper consists of 19 printed pages and 1 blank page.

1. Radioactive amino acids are supplied to a cell that uses them to make insulin.



Which route will the radioactive amino acids take?

	First				Last
A	4	2	3	1	5
B	4	3	2	1	5
C	5	1	3	2	4
D	5	3	2	4	1

2. A certain cell surface membrane is made up entirely of phospholipids and is 5nm thick. A volume of 1 mm^3 of this tray of phospholipids membrane was homogenized and dropped onto the surface of water in a large tray. The phospholipids spread out to form a continuous thin film.

   What is the expected area of this film?

   A 200 000 mm^2 because the molecules form a single layer
   B 200 000 mm^2 because the molecules form a double layer
   C 400 000 mm^2 because the molecules form a single layer
   D 400 000 mm^2 because the molecules form a double layer

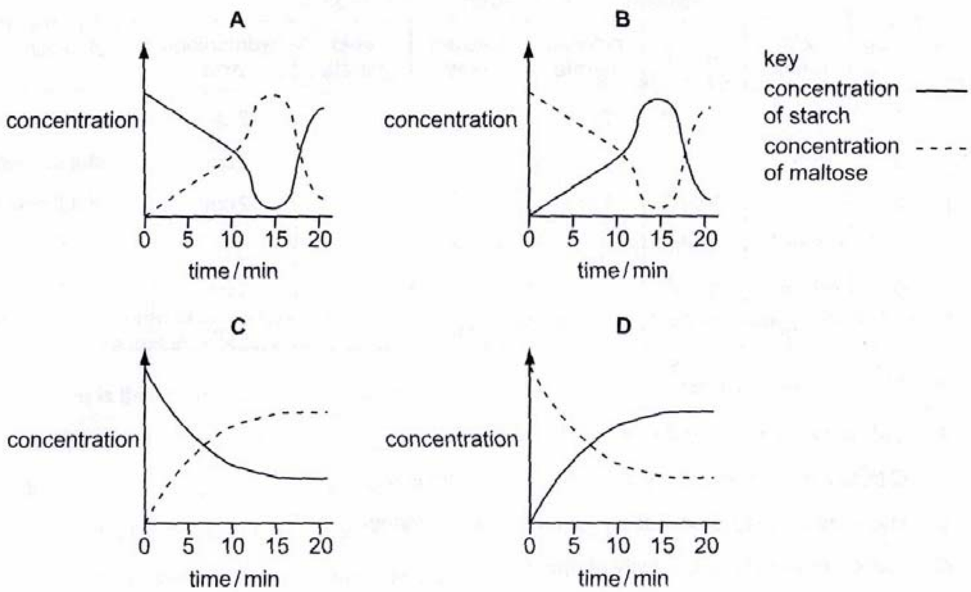
3. The diagram shows the 3D conformation of a protein.

What would happen when this protein denatures?

A It will be hydrolyzed into its component amino acids.
B All the α-helices and β-pleated conformation will be lost.
C It loses its secondary and tertiary structures.
D Its hydrogen bonds, ionic bonds, hydrophobic interactions, disulphide linkages as well as peptide bonds are disrupted.

4. In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. 1% hydrochloric acid is added after 10 minutes and the mixture is heated to 60ºC at 14 minutes.

   Which graph represents the results of this experiment?



5. The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

Which one of the following statements is **not** consistent with these results?

A	At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
B	At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
C	In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
D	At a light intensity of 10% of full sunlight, photosynthesis does not occur.

6. Some photosynthetic organisms containing chloroplasts that lack PS II (photosystem II) are able to survive. The best way to detect the lack of PS II in these organisms would be to;

A determine if they have thylakoids in the chloroplasts.
B test for liberation of oxygen in the light.
C test for carbon fixation in the dark.
D determine the production of starch.

7. The diagram summarizes the pathway of glucose breakdown.

$$\begin{array}{c} \text{Hexose} \\ \downarrow \mathbf{1} \\ \text{Triose phosphate} \\ \downarrow \mathbf{2} \\ \text{Pyruvate} \\ \downarrow \\ \text{Acetyl CoA} \\ \downarrow \mathbf{3} \\ \text{6C compound} \\ \downarrow \mathbf{4} \\ H_2O + CO_2 \end{array}$$

Which two steps results in a net increase of ATP?

A 1 and 3.
B 1 and 4.
C 2 and 3.
D 2 and 4.

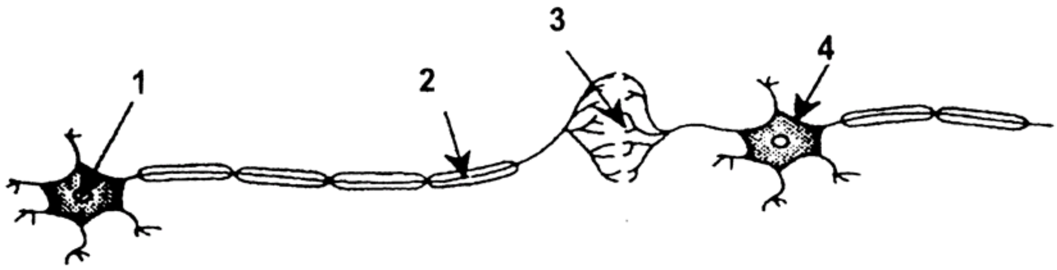
8. The events shown below occur during different phases of mitosis.

1. Spiralisation of DNA
2. Reappearance of nuclear membrane
3. Centromeres split
4. Centromeres attach to spindle fibres
5. DNA replicates

Which of the following correctly identifies each of the phases described?

	Interphase	Prophase	Metaphase	Anaphase	Telophase
A	1	2	3	4	5
B	5	1	4	3	2
C	3	2	4	1	5
D	4	5	1	2	3

9. The diagram below represents an impulse pathway.



Nerve gas interferes with the action of an enzyme that breaks down acetylcholine.

At which location does this inhibiting effect of the nerve gas occur?

A 1
B 2
C 3
D 4

10. Which of the following can activate a protein by transferring a phosphate group to it?

1 cAMP
2 phosphodiesterase
3 phosphotransferase
4 protein kinase

A 1 and 3
B 2 and 3
C 2 and 4
D 3 and 4

11. Which of the following statements is **true** regarding point mutations?

1 They could result in a frameshift mutation.
2 They always produce a change in the amino acid sequence of a protein.
3 They could result in genes being duplicated.

A 1 only
B 1 and 2
C 2 and 3
D 1, 2 and 3

12. A new form of life is discovered. It has a genetic code much like that of organisms on Earth except that there are **five** different DNA bases instead of four and the base sequences are translated as **doublets** instead of triplets. How many different amino acids could be accommodated by this genetic code?

A 25
B 32
C 10
D 64

13. The table below shows some of the events which take place in protein synthesis.

1 tRNA molecules bring specific amino acids to the ribosome.
2 Ribonucleotides join with exposed DNA bases and form a molecule of mRNA.
3 Release factor binds to mRNA.
4 Peptide bonds formed between amino acids.
5 mRNA molecule leaves the nucleus.
6 mRNA undergoes capping and addition of poly(A) tail.

Which of the following shows the correct sequence of events?

A 2 5 6 1 4 3
B 2 6 5 1 4 3
C 2 6 5 1 3 4
D 2 5 6 1 3 4

14. A biochemist isolates and purifies various molecules needed for DNA replication. When she adds some DNA, replication occurs but the DNA molecules formed are defective. Each consists of a normal DNA strand paired with numerous segments of DNA a few hundred nucleotides long. What has she probably left out of the mixture?

A DNA polymerase
B nucleotides
C primers
D ligase

Use the information given below to answer Questions 15 and 16.

Shown in **Fig. 15** is the genomic structure of the wild-type human β-globin gene. The numbers within the boxes indicate the length of nucleotides of each region, **inclusive** of bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codons are indicated.

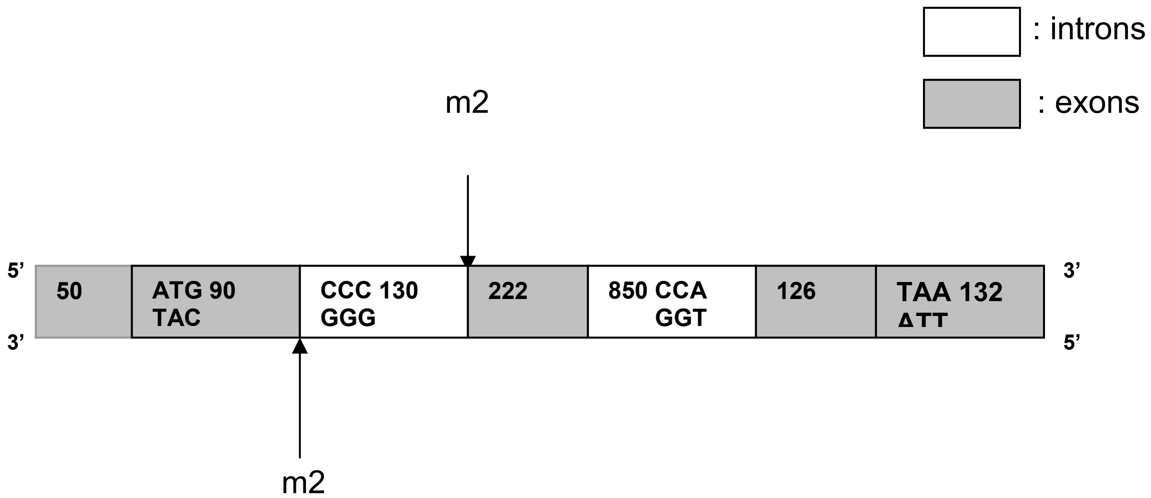


Fig. 15

15. What is the length (in nucleotides) of the wild-type β-globin primary RNA transcript and how many amino acids are present in the wild type β-globin protein?

	Length of β -globin primary RNA transcript	**No. of amino acid in wild-type β – globin protein**
A	1600	146
B	620	146
C	1600	206
D	620	206

16. A base-pair substitution mutations (m2) indicated above, destroys the 3' splice site located in the first intron of the β-globin gene. The splice site is the site where the DNA will be cut by enzymes.

Choose the correct explanation as to whether the m2 mutation would any change in the length of the primary RNA transcript and mature RNA transcript made from the mutant β-globin gene.

A	**Pre-mRNA:**	No change as this is before the mRNA is spliced.
	Mature mRNA:	Because the splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
B	**Pre-mRNA:**	No change as this is before the mRNA is spliced.
	Mature mRNA:	Because the splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.
C	**Pre-mRNA:**	An increase in its length as this is before the mRNA is spliced.
	Mature mRNA:	Because the splice site is damaged, the intron remains and the mature mRNA is increased by 130 bases.
D	**Pre-mRNA:**	A decrease in its length as this is before the mRNA is spliced.
	Mature mRNA:	Because the splice site is damaged, the exon remains and the mature mRNA is increased by 222 bases.

17. Identify the **CORRECT** statement.

A A promoter sequence region (for transcription) in a bacterial cell is normally transcribed into the primary transcript.

B In mammalian cells, promoter sequences are normally located at the 3' end of a gene.

C In mammalian cells, transcribed sequences never participate in promoter function.

D In bacterial cells, the promoter region is normally located at the 5' end of a gene.

18. Enhancers of gene expression are;

A proteins that regulate spatial and temporal control of genes.

B regulatory DNA sequences.

C used for expression of all genes.

D never found in eukaryotes.

19. Which of the following is **not** true of telomerase?

A Telomerase contains a ribozyme.
B Telomerase activity decreases as the cell ages.
C Telomerase adds a repeating sequence to the 5' end of DNA.
D Telomerase adds a repeating sequence to the 3' end of DNA.

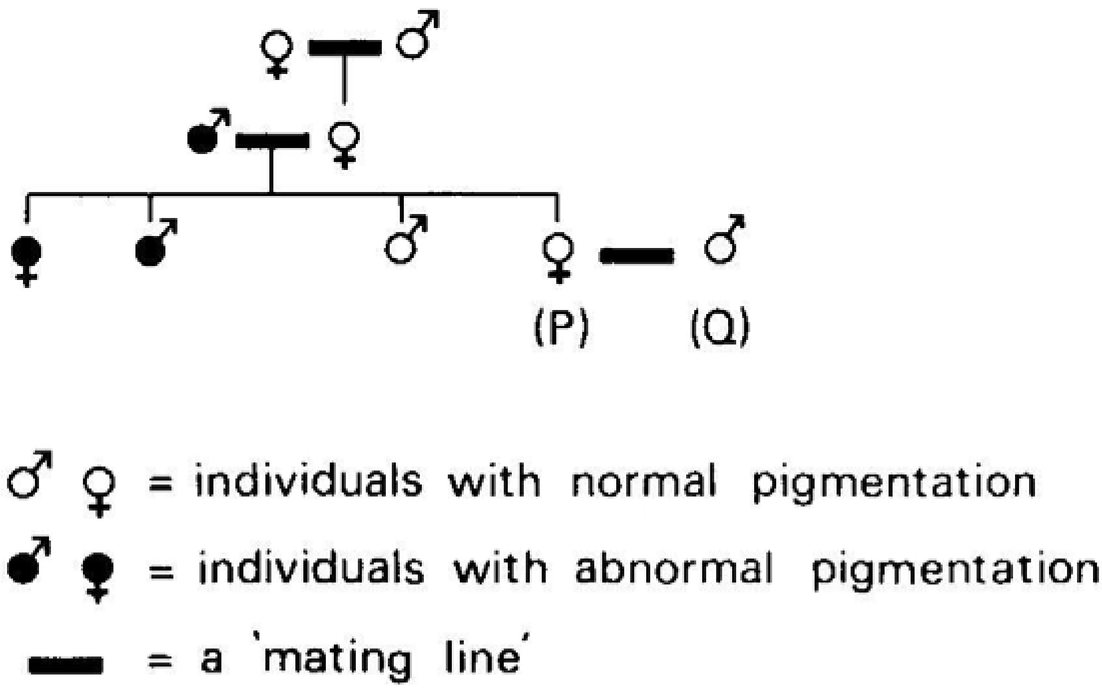
20. The table shows a comparison between the genomes of a prokaryote and eukaryote.

	Prokaryote	**Eukaryote**
Genome length (base pairs)	4 640 000	12 068 000
Number of proteins	4300	6200
Proteins with roles in metabolism	650	650
Proteins with roles in energy release	240	175
Proteins with roles in protein synthesis	410	750

Which feature of eukaryotes accounts for the differences in the number of proteins with roles in protein synthesis?

A The DNA of a eukaryote has histones.
B The DNA of a eukaryote has introns.
C The DNA of a eukaryote has more base pairs.
D The DNA of a eukaryote is separated from the ribosomes by membranes.

21. In a species where the female is homogametic, a sex-linked allelic pair of genes controls pigmentation. The following results were obtained during the course of a breeding experiment.



Which of the following ratios of offspring will be produced when P and Q are bred together?

A 2 females (both normal) : 2 males (both abnormal)
B 2 females (both carriers) : 2 males (both normal)
C 2 females (1 carrier, 1 abnormal) : 2 males (1 normal, 1 abnormal)
D 2 females (1 carrier, 1 normal) : 2 males (1 normal, 1 abnormal)

22. The condition in which a zygote contains three copies of a particular chromosome as a result of nondisjunction is called _____.

A trisomy

B monosomy

C polyploidy

D monoploidy

23. The table below shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1st parent	Genotype of 2nd parent	
	$I^A I^A$	$I^A I^O$
$I^A I^O$	A	A and O
$I^B I^O$	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria.

What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

A 1 in 16
B 1 in 8
C 3 in 16
D 3 in 8

24. In pea-plants, flower color and height are determined by two pairs of alleles. The genes for red color and tallness are dominant. Two flowers were crossed, and the offspring (F1) were self-pollinated.

The results in the F2 generation were as follows: 280 tall red; 90 tall white; 94 dwarf red; 30 dwarf white. Which one of the following is a possible description of the **parental generation**?

A heterozygous tall red x heterozygous dwarf white

B homozygous tall red x homozygous dwarf white

C heterozygous tall red x homozygous dwarf white

D homozygous tall white x heterozygous dwarf red

25. In domestic poultry, the gene for white feathers is dominant to the gene for dark feathers and the gene for frizzled feathers is dominant to normal feathers. Poultry which were homozygous for dark, frizzled feathers were bred with poultry bearing white, normal feathers. Members of the F_1 were test corssed with homozygous recessives and produced the following offspring;

White frizzled	18
Dark frizzled	63
White normal	63
Dark normal	13

These results are due to;

A sex-linkage

B non-disjunction

C Mendalian dihybrid inheritance

D linkage and chiasmata inheritance

26. In a species of plant, the allele for hairy stems is dominant to the allele for smooth stems. The allele for purple stems is dominant to the allele for green stems.

A plant was self-fertilized and the offspring appeared in the ratio;

9 hairy, purple stem: 3 hairy, green stem: 3 smooth, purple stem: 1 smooth, green stem.

The offspring with smooth, green stems were crossed to all the other offspring.

What proportion of these crosses would be expected to produce offspring in the ratio 1 hairy, purple stem: 1 hairy, green stem: 1 smooth, purple stem: 1 smooth, green stem?

A 9/15

B 6/15

C 4/15

D 3/15

27. A virus has a base ratio of (A + G) / (U + C) = 1. What type of virus is this?

A a single-stranded DNA virus
B a double-stranded DNA virus
C a single-stranded RNA virus
D a double-stranded RNA virus

28. The hepatitis C virus is a single-stranded, enveloped, positive sense RNA virus, which attack the liver cells of human beings.

Which of the following general strategy state how the viruses enter the host cell?

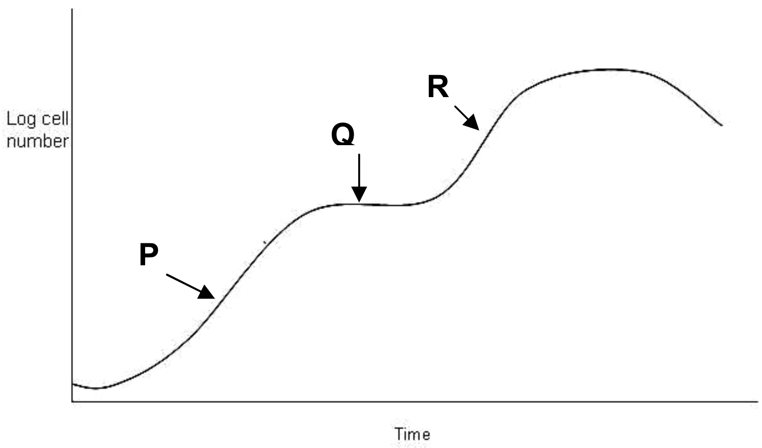
A uncoating of the genetic material of the virus via endocytosis

B fusion of the viral envelope with the host cell membrane

C injection of the whole virus into the host cell

D penetration of the viral capsid via budding

29. *E.coli* bacteria are grown in a culture of nutrients, which includes glucose and lactose as the main source of carbon-based nutrient.



Which of the following statements accounts for the growth curve showing the reproduction of *E.coli* bacteria?

	P	Q	R
A	Lactose is metabolised	Production of enzymes for lactose metabolism is repressed	Glucose is metabolised
B	Lactose is metabolised	Enzymes for lactose metabolism is produced	Glucose is metabolized
C	Glucose is metabolised	Production of enzymes for lactose metabolism is repressed	Lactose is metabolised
D	Glucose is metabolised	Enzymes for lactose metabolism is produced	Lactose is metabolised

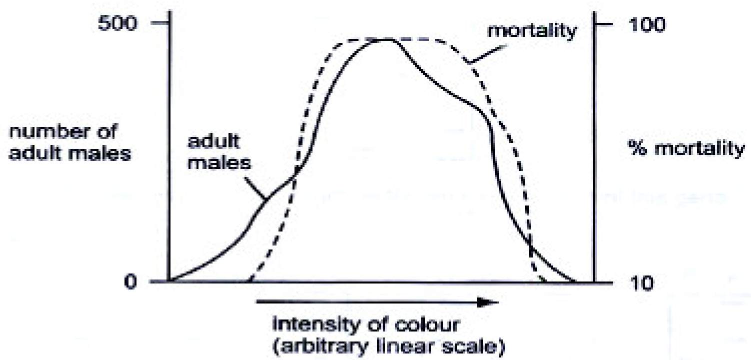
30. Some events that take place during generalized transduction are listed below.

I	Bacterial host DNA is fragmented.
II	Bacterial DNA may be packaged in a phage capsid.
III	Recombination between donor DNA and recipient DNA.
IV	Phage infects bacterial cell.
V	Phage DNA and proteins are made.

Which sequence of events is correct?

	First				Last
A	IV	I	III	V	II
B	IV	I	V	II	III
C	IV	III	I	V	II
D	IV	V	I	III	II

31. The graph below shows data on a population of a species of moth which shows considerable variation in color intensity.



Which conclusion can be made from this graph?

A Color variation is environmentally induced.
B Color variation is genetically determined.
C Extreme forms are favored by natural selection.
D The species shows discontinuous variation with respect to color.

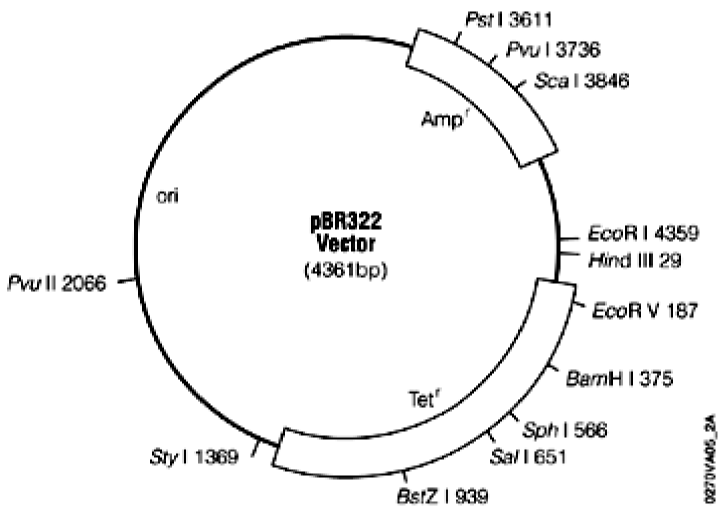
32. Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A The fossil records.
B Homologous structures.
C Comparative embryology.
D Molecular comparisons of DNA and amino acid sequences.

33. What are the major effects of gene flow between subpopulations?

A Increase genetic variation within and between subpopulations.
B Decrease genetic variation within and between subpopulations.
C Decrease genetic variation within but increase genetic variation between subpopulations.
D Increase genetic variation within but decrease genetic variation between subpopulations.

34. pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease *Bam*H1.



Following transformation, bacterial cells were grown in four different media as shown below;

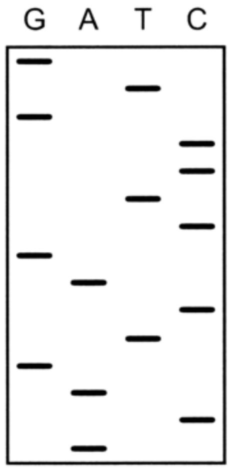
I	Nutrient broth plus ampicillin
II	Nutrient broth plus tetracycline
III	Nutrient broth plus ampicillin & tetracycline
IV	Nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

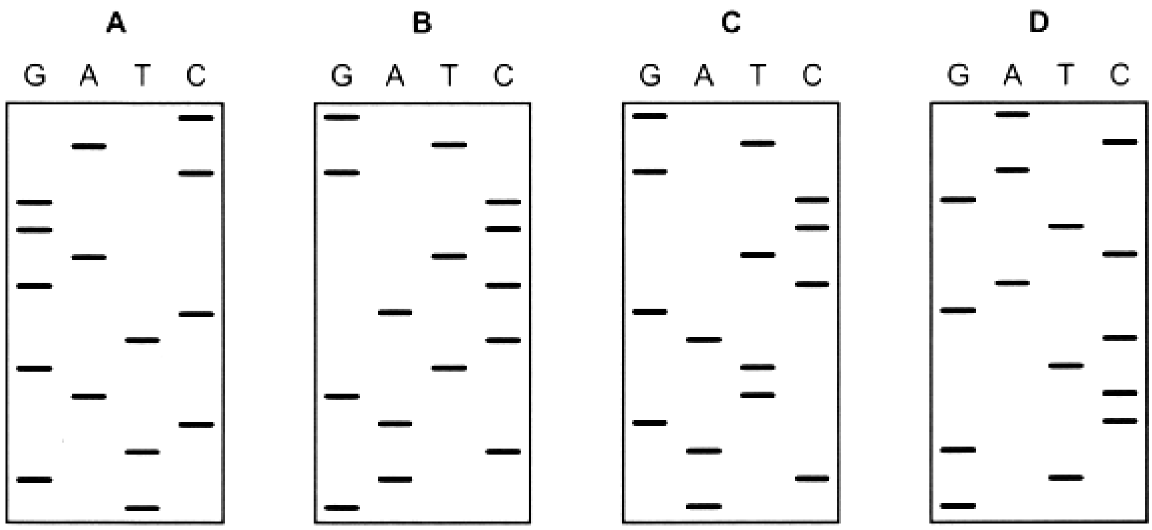
A I and II
B I and III
C I and IV
D IV only

35. In people with a mutant allele, a protein contains just one different amino acid in its primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using gel electrophoresis.

The gel electrophoresis result from the DNA of a normal allele of a gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



36. A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

(i) Transfer of DNA to nitrocellulose membrane.
(ii) Restriction digestion of genomic DNA.
(iii) Cleaved DNA separated using gel electrophoresis.
(iv) Create radioactive probe.
(v) Incubate probe and membrane.

Which is the correct sequence to the above steps?

A	ii → iii → i → iv → v.
B	ii → iii → i → v → iv.
C	iv → v → i → ii → iii.
D	v → iv → iii → ii → i.

37. Nuclei from both the liver and brain cells of a rat are isolated and incubated with highly radioactive RNA precursors. The RNA transcripts synthesised in these nuclei become radioactively labeled.
These radioactive RNA molecules are incubated with a single-stranded DNA segment (cDNA) complementary to an mRNA molecule found in liver-cell cytoplasm but not in brain-cell cytoplasm. The RNA and DNA are allowed to form RNA/DNA hybrid double helices with the specific cDNA probe.
An RNA-digesting enzyme is added to destroy RNA that is not in hybrid form.
Radioactivity in the remaining hybrids is measured as counts per minute.

The results are presented in the table below.

Incubation components	Counts per minute
Liver cDNA probe plus primary RNA transcripts from liver-cell nuclei	15,000
Liver cDNA probe plus primary RNA transcripts from brain-cell nuclei	150

Which of the following statements concerning the DNA sequence corresponding to this liver-specific mRNA molecule is best supported by the data?

A It is transcribed in both brain- and liver-cell nuclei at approximately the same rate.
B It is not transcribed in either liver- or brain-cell nuclei.
C It is not transcribed in brain-cell nuclei but is transcribed in liver-cell nuclei.
D It may be transcribed in both brain- and liver-cell nuclei but cannot be detected by this experimental procedure.

38. What are the normal functions of stem cells in a living human?

	Functions		
A	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing insulin in pancreas damaged by type 1 diabetes	cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing red blood cells worn out by normal wear and tear	cells that can differentiate into cardiac muscles in the heart
C	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing dopamine in brains of people with Parkinson's disease	cells that can differentiate into cartilage cells in the joint
D	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing differentiated cells that can be used to screen new drugs	cells that can differentiate into nerve cells in the brain

39. A patient suffering from cystic fibrosis underwent gene therapy using a viral mediated delivery system. The treatment failed and you were asked to troubleshoot the problem.

Which of the following could not be the explanation behind the failure in the treatment?

A Failure of the expressed protein to be folded into the correct conformation.
B Integration of the target gene in the enhancer region.
C Rejection of the vector by the host immune system.
D The CFTR protein was not expressed in adequate amount.

40. Plants are more readily manipulated by genetic engineering than are animal cells because;

A plant genes do not contain introns.
B more vectors are available for transferring recombinant DNA into plant cells.
C a somatic plant cell can often give rise to a complete plant.
D plant cells have larger nuclei.